DESIGN AND FABRICATION OF MULTI SPINDLE DRILLING MACHINE

MEB1204 - MANUFACTURING TECHNOLOGY- II

Submitted by

KARTHICK S 8115U24ME029

in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

In

MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(AUTONOMOUS)

SAMAYAPURAM, TRICHY

ANNA UNIVERSITY

CHENNAI 600 025

DECEMBER-2025

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Mr. S. SIVANANTHAN., M.E., (Ph.D.)

Department of Mechanical Engineering

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

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(AUTONOMOUS)

Under

ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

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Under

ANNA UNIVERSITY, CHENNAI

DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva- Voce held at K. Ramakrishnan College of Engineering on _____

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SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

We thank the Almighty God, for showing abundance of grace, without his blessings it would not have been possible for us to complete our project.

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We would like to thank our parents and friends for their constant support and encouragement throughout this project.

INSTITUTE VISION AND MISSION

VISION

- To achieve a prominent position among the top technical institutions

MISSION

- To bestow standard technical education par excellence through state-of-the-art infrastructure, competent faculty and high ethical standards.
- To nurture research and entrepreneurial skills among students in cutting edge technologies.
- To provide education for developing high-quality professionals to transform the society.

DEPARTMENT VISION AND MISSION

VISION

To provide eminent mechanical engineering to the industries and society through quality education.

MISSION

M1. To impart quality professional education through state-of-the-art infrastructures.

M2. To motivate students and faculty to undertake research and development activities in the thrust areas of mechanical engineering.

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PSO1: Apply the logical, analytical and technical skills to model and build electrical systems and appliances as per societal requirements.

PSO2: Apply the advanced and fundamentals Electrical and allied Engineering knowledge in the design and development of hardware and software tools for non- conventional electrical power generation and distribution.

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PEO1: Have strong foundation in Electrical and Electronics Engineering to excel in professional career, in higher studies or research.

PEO2: Analyze, design and develop various interdisciplinary projects and products, to solve social issues.

PEO3: Have professional ethics and effective communication skills with life-long learning attitudes.

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PO1 Engineering knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2 Problem analysis Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3 Design/development of solutions Design creative solutions for complex

engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).

PO4 Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5 Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6 The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7 Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8 Individual and Collaborative team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

PO9 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to

one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context

ABSTRACT

A multi-spindle drilling machine is designed to perform simultaneous drilling operations on a workpiece, significantly increasing productivity and accuracy compared to conventional single-spindle machines. This project focuses on developing a cost-effective and efficient multi-spindle system capable of drilling multiple holes at predetermined positions in a single cycle. The design integrates a power transmission mechanism, adjustable spindle spacing, and a stable fixture to ensure uniform hole size, improved repeatability, and reduced machining time. By minimizing manual effort and cycle time, the multi-spindle drilling machine enhances overall manufacturing efficiency, making it suitable for mass production industries such as automotive, fabrication, and small-scale workshops. The project demonstrates that multi-spindle drilling technology can offer high precision, operational safety, and improved production rates while maintaining economic feasibility.

Keywords: Power transmission mechanism, Adjustable Spindle Centre distance, Feed rate and cutting speed and Fixture and clamping system.

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CHAPTER 1

(INTRODUCTION)

Manufacturing industries across the world continually seek innovations that improve efficiency, precision, and productivity while reducing operational costs and human effort. Drilling is one of the most fundamental machining processes used in nearly every sector of engineering production, ranging from automotive and aerospace to fabrication workshops and small-scale industries. Traditional single-spindle drilling machines, though widely used, often fall short when large quantities of components with multiple holes need to be produced. Repetitive operations, frequent repositioning of the workpiece, and the need for skilled labor increase production time and introduce the possibility of human error. These limitations highlight the growing need for more advanced drilling solutions that can handle high-volume, precision-demanding tasks with greater consistency and reduced cycle time. In response to these industrial demands, the concept of a multi-spindle drilling machine has emerged as an effective and efficient alternative.

A multi-spindle drilling machine is a specialized system designed to perform multiple drilling operations simultaneously by using two or more spindles driven by a common power source. The simultaneous drilling capability drastically reduces machining time while ensuring uniformity in hole dimensions and placement. Instead of performing each drilling operation separately, the multi-spindle system enables mass production with improved accuracy and reduced operator fatigue. The introduction of this technology has significantly transformed production environments where speed and consistency are crucial. Industries that require symmetrical hole patterns, repeated drilling layouts, and

large-scale component batches greatly benefit from the implementation of multi-spindle drilling systems.

The motivation behind developing a multi-spindle drilling machine lies in its ability to address the shortcomings of conventional machines. While single-spindle systems are suitable for small-scale or low-volume tasks, they lack the efficiency needed for industries running continuous production lines. Multi-spindle drilling machines not only reduce cycle time but also enhance precision by eliminating errors associated with manual repositioning. Furthermore, they improve tool utilization, reduce material handling, and ensure consistent performance across large batches of products. By increasing output without compromising on quality, such systems serve as an ideal solution for modern manufacturing challenges.

The increasing emphasis on automation and productivity has sparked interest in machines that deliver optimal performance with minimal human intervention. Multi-spindle drilling machines align with these industrial expectations by integrating mechanisms that simplify operations while maintaining robustness and accuracy. Their ability to handle various materials, including metals, plastics, and composite components, further expands their applicability across different sectors. Whether in mass production of automotive parts, fabrication of mechanical components, or assembly operations requiring precision drilling, the versatility of multi-spindle systems makes them indispensable in today's competitive market.

The development of this machine involves a comprehensive understanding of mechanical design, power transmission systems, spindle alignment, material selection, and safety features. Proper alignment of multiple spindles is essential to achieve accurate hole positioning, while the power transmission system must ensure smooth and balanced torque distribution to each spindle. The structural

design must be strong enough to withstand high-speed operations without causing vibrations that could affect drilling quality. Additionally, operator safety and ease of maintenance must be incorporated into the design to ensure long-term usability and reliability. By considering these technical aspects, the project aims to produce a machine that is not only efficient but also durable and user-friendly.

In essence, this project seeks to explore, design, and develop a multi-spindle drilling machine that addresses the evolving needs of the manufacturing industry. As production technologies advance and industries demand higher throughput rates, machines capable of delivering rapid, precise, and cost-effective drilling solutions become increasingly valuable. The introduction of a multi-spindle drilling system represents a step toward enhancing industrial productivity while maintaining high standards of quality and safety. This project aims to contribute meaningfully to the field of machining technology by presenting a machine that embodies efficiency, reliability, and innovation in drilling operations.

CHAPTER 2

LITERATURE SURVEY

(2025) Manoj et al. - Shreeyash Manoj et al. analyzed spur gear transmissions in drilling/tapping SPMs, ensuring reliable power distribution for precise multi-spindle operations

(2024) Deshmukh et al. - Deshmukh et al. built mountable attachments for conventional machines, achieving time reductions for flange components through easy setup and dismantling.

(2022) Kumar et al. - Kumar et al. designed lightweight portable multi-spindle heads for on-site use, detailing gear boxes that enable precise drilling where large machines cannot access.

(2021) Raut et al. - Shreyank Prakash Raut et al. extended multi-spindle concepts to multi-axis setups, reducing setups for complex components via controller-based operations.

(2021) Choudhary et al. - Mahendra O. Choudhary et al. designed an automated multi-spindle setup with gearbox for high-speed drilling on square shafts, focusing on gear selection and force analysis to achieve maximum accuracy and production rate. Their SolidWorks modeling showed reduced labor costs and cycle times without quality degradation .

(2020) Kolgiri et al. - Somnath Kolgiri et al. analyzed adjustable multi-spindle heads with sliding carriers and gears, evaluating power efficiency, speed uniformity, and flexibility for varied hole patterns .

(2020) Gawande and Trikal - S.R. Gawande and S.P. Trikal studied attachments increasing productivity via accurate multi-hole drilling, concluding reduced cycle times and better performance on existing machines .

(2019) Rai - Gaurav Rai developed a low-cost multi-spindle machine using geared heads to drill multiple holes simultaneously, emphasizing productivity gains for small industries through simple mechanical designs .

(2019) Singh et al. - Singh et al. optimized multi-spindle SPM for engine hoods in mass production, focusing on low-variety jobs with high hole counts and consistent accuracy .increases via fixed or adjustable center designs in mass production .

(2018) Shingavi et al. - Prof. Ms. A.A. Shingavi et al. optimized multi-spindle heads for cycle time reduction, highlighting positional accuracy and productivity increases via fixed or adjustable center designs in mass production .

(2018) Bui and Tran - Bui Ngoc Tam and Tran Van Dich prototyped a vertical multi-spindle machine for small-batch use, validating structure, transmission, and accuracy through experiments showing improved industrial productivity .

(2017) Rakate - G.N. Rakate fabricated a varying-center multi-spindle attachment, noting equal drilling time for multiple holes and enhanced positional accuracy over single-spindle setups .

(2017) Patel et al. - Vinay D. Patel et al. used ANSYS to validate stresses in multi-spindle heads, confirming safe designs for simultaneous dual-hole drilling with variable centers .

(2016) Pawar et al. - Pawar et al. reviewed SPM designs across industries, comparing fixed/adjustable heads and recommending CNC integration for future automation and accuracy .

(2015) Jadhav et al. - Jadhav et al. created interchangeable multi-spindle heads mountable on conventional drills, demonstrating substantial productivity boosts and uniform drilling time for multiple holes .

CHAPTER-3

PROBLEM IDENTIFICATION

1.Low Productivity in Single-Spindle Drilling

Traditional single-spindle machines can drill only one hole at a time, resulting in high machining time.

2. Increased Cycle Time

Multiple drilling operations on a single workpiece require repeated repositioning, increasing cycle time.

3. Inconsistent Hole Accuracy

Manual positioning often leads to variations in hole spacing, diameter, and alignment.

4. High Labour Effort

Operators must manually change positions and tools, leading to fatigue and reduced efficiency.

5. Difficulty in Maintaining Repeatability

Producing identical hole patterns consistently is challenging with conventional machines.

6. Lack of Synchronisation in Multiple Drilling Operations

Performing holes one-by-one may create non-uniform machining forces and inconsistent surface quality.

7. Higher Production Cost

Increased machining time, manpower requirements, and low repeatability raise overall production cost.

8. Limited Suitability for Mass Production

Single-spindle machines fail to meet the demand for high-volume manufacturing industries.

9. Tool Wear Due to Uneven Cutting Conditions

Inconsistent feed and drilling parameters cause premature tool wear and poor surface finish.

10. Complex Setup Process

Repeated alignment and clamping of the workpiece for multiple holes increase setup time and error probability.

In modern manufacturing industries, drilling operations form a critical stage in producing components with precise holes required for assembly, fastening, and structural integrity. Traditional single-spindle drilling machines, although reliable, present several limitations when used in high-volume production environments. The most significant problems arise from their inability to perform multiple drilling operations simultaneously, leading to increased machining time, reduced productivity, and higher operational costs. These limitations become more severe in industries where large batches of identical components must be produced within strict deadlines.

A major issue associated with conventional drilling systems is the excessive cycle time. Since only one hole can be drilled at a time, operators must repeatedly reposition the workpiece to complete multiple holes. This not only extends the total manufacturing time but also increases operator fatigue and the likelihood of human error. Misalignments during repositioning can cause dimensional inaccuracies, compromising the quality of the final component. As production targets increase, maintaining consistent accuracy with single-spindle machines becomes increasingly challenging.

Another problem lies in the lack of uniformity in multi-hole drilling operations. When each hole is drilled separately, slight variations in feed rate, tool wear, and operator handling can lead to inconsistencies in hole diameter, surface finish, and hole spacing. These variations can negatively impact assembly processes and the performance of the final product. In precision-dependent industries, even minor deviations from design specifications can result in component rejection, increasing material wastage and production costs.

Furthermore, single-spindle machines contribute to higher labour involvement. Operators must constantly monitor the machine, adjust tool positioning, and ensure accuracy for each individual hole. This limits automation potential and increases dependency on skilled labour, which may not always be available in small- and medium-scale industries. The repetitive nature of the process also reduces overall efficiency and increases operational fatigue.

The rigidity and structural limitations of conventional drilling setups also become evident in mass-production applications. Machines designed for single-point operations often lack the mechanical strength and stability required for simultaneous multi-hole drilling, constraining speed, feed, and tool life. As a result, industries struggle to balance productivity with tool durability and surface finish quality.

Additionally, production facilities with multiple single-spindle machines face issues related to space utilization and energy consumption. Operating several machines to achieve higher output increases electricity usage, maintenance requirements, and shop-floor congestion. This setup is far from optimal in terms of cost-efficiency and workflow organization.

Therefore, the need for a more efficient, accurate, and cost-effective drilling solution becomes evident. The identification of these problems highlights the demand for a multi-spindle drilling machine capable of drilling multiple holes simultaneously with consistent accuracy. Such a system would address the delays, inconsistencies, labour dependency, and high operational costs currently experienced with traditional methods, thereby enhancing overall productivity and manufacturing quality

CHAPTER 4

OBJECTIVE AND SCOPE OF THE PROJECT

The main objective of this project is to design and develop a Multi-Spindle Drilling Attachment that improves productivity, enhances drilling accuracy, and reduces operational time in manufacturing environments. The system aims to enable simultaneous drilling of multiple holes, ensuring uniform spacing and consistent depth while minimizing human effort.

Specific Objectives:

1.To reduce overall machining time by performing multi-hole drilling in a single operation.

2.To improve accuracy and repeatability in drilling operations.

3.To design a compact and cost-effective attachment suitable for conventional lathe or drill machines.

4.To enhance operator safety by minimizing manual alignment and repetitive tasks.

5.To evaluate performance through speed, accuracy, and efficiency improvements compared to single-spindle drilling.

Scope of the Objective

This objective focuses on increasing production efficiency in small and medium-scale industries by implementing a mechanically driven multi-spindle attachment. The design will consider factors such as spindle alignment, power transmission, material selection, manufacturing feasibility, and cost optimization. Performance metrics will include drilling time reduction, hole accuracy, tool wear comparison, and stability during operation.

The project will ultimately offer a practical solution for industries requiring repetitive drilling operations with high precision and reduced cost.

SCOPE OF THE PROJECT

The scope of this project centers on the development and practical implementation of a Multi-Spindle Drilling Attachment designed to enhance productivity and accuracy in repetitive drilling operations. This scope outlines the boundaries, functional expectations, and application areas related to achieving the project's objectives. The multi-spindle attachment will focus on enabling simultaneous drilling of multiple holes with consistent spacing, uniform depth, and improved precision compared to traditional single-spindle drilling methods. The project encompasses the conceptual design, detailed engineering analysis, prototype development, and performance evaluation of the attachment.

The scope includes studying spindle configurations, power transmission mechanisms, gear arrangements, and material selection to ensure durability and operational stability. It also covers the integration of the attachment with existing conventional drilling or lathe machines, emphasizing user compatibility and ease of installation. Considerations such as vibration control, spindle alignment, machining parameters, and tool wear behavior will be studied to ensure optimal performance.

Additionally, the scope extends to conducting comparative analysis between single-spindle and multi-spindle drilling to measure improvements in machining time, hole accuracy, repeatability, and operator effort. This comparison will help validate the effectiveness of the developed system. The project will be limited to mechanical methods of power transmission and will not involve advanced automation systems like CNC or PLC control. It will focus on applications suitable for small-scale and medium-scale industries where cost-effective productivity enhancement tools are critical.

Overall, the scope ensures that the multi-spindle drilling attachment is designed to be practical, economical, safe, and efficient—providing industries with a reliable solution for high-precision repetitive drilling tasks.

CHAPTER 5

MATERIALS AND METHODS

Materials:

Frame material

Mild steel (MS) square/rectangular tube and flat bar, used for base and vertical column. Typical size: MS square tube 25 × 25 × 3 mm; MS flat 40 × 6 mm.

Base and work support plate MS plate for rigid base frame and workpiece support. Typical thickness: 6–8 mm for base, 5–6 mm for work plate.

Guide column and slide Round MS/EN8 steel rod used as guide column for vertical movement of the drill holder. Diameter about 25–30 mm; length according to required stroke.

Rack-and-pinion / feed mechanism MS rack welded or fixed to guide column, with matching pinion gear mounted on a shaft with operating lever. Simple hand lever of MS flat or pipe to give manual feed.

Drill holder and multi-spindle head body Solid MS block or plate to hold bearings and small spindles. Thickness about 20–30 mm depending on number of spindles.

Spindles and bushes Medium carbon steel (EN8/EN19) rods for individual drilling spindles. Bronze / gunmetal bushes or ball bearings for smooth rotation.

Power drill (hand drill) Commercial portable electric drill, single-phase, about 500–800 W, mounted on the vertical slide. Chuck capacity typically 10–13 mm.

Gears / pulleys for multi-spindle drive Small spur gears or bevel gears (hardened steel) used inside the head to transmit motion from main drill to each spindle. Gear module and tooth numbers selected to keep equal speed on all spindles.

Fasteners and accessories Standard nuts, bolts, washers (M8, M10, M12) for frame assembly and mounting of drill, head, and bearings. Grease, cutting oil, and electrical on/off switch with cable for safe operation.

ITEM	SPECIFICATION / NOTES
Mild Steel (MS)	Used for frame, base, plates; typical sizes: 25×25×3 mm tubes, 40×6 mm flats, 6–8 mm plates.
EN8 Steel	Medium-carbon steel rods used for guide columns (Ø25–30 mm) and spindles.
EN19 Steel	High-strength alloy steel used for heavy-duty spindles.
Bronze	Used for bushes providing smooth low-friction rotation.
Gunmetal	Alternative bush material offering good wear resistance
Hardened Steel	Used for gears inside the multi-spindle head for durability
Spur Gears	Small steel gears used for equal-speed power transmission between spindles.
Bevel Gears	Used to change direction of rotation inside the multi-spindle head
Ball Bearings	Support spindles for smooth, high-speed rotation
Fasteners (Nuts, Bolts, Washers)	Standard M8/M10/M12 hardware for assembly.
Grease	Lubricant for bushes, gears, and bearings.
Cutting Oil	Used during drilling to reduce heat and improve tool life.

The multi spindle drilling machine consists mainly of a welded mild steel frame made from square or rectangular tubes and flat bars, providing a rigid base and vertical support for all components. A solid mild steel plate forms the work support at the bottom, while a hardened or medium-carbon steel round bar acts as the vertical guide column on which the drill carriage slides. The feed mechanism uses a steel rack fixed to the column and a matching steel pinion mounted on a shaft with a lever made from steel flat or pipe to apply manual feed. The multi-spindle head body is machined from a thick mild steel block that houses medium-carbon steel spindles supported in bushes or ball bearings, and spur or bevel gears made from alloy or hardened steel distribute motion from the main drive to each spindle. A commercially available single-phase portable electric hand drill with a 10–13 mm capacity chuck serves as the power source and is clamped to the sliding carriage. Standard steel nuts, bolts, and washers are used for assembly, while lubricants and cutting fluids ensure smooth motion and improved tool life.

DESIGN AND LAYOUT

A simple stand-type layout was prepared with a rectangular base frame, vertical column, and sliding carriage to hold the hand drill and multi-spindle head. Spacing between spindles was fixed according to the required hole pitch in the workpiece.

Fabrication of frame and base

MS tubes and flats were cut to size and welded to form the base rectangle and vertical support, similar to the photo. A work support plate was welded or bolted near the bottom to keep the job under the spindles.

Guide column and sliding carriage

A round steel column was fixed to the frame; a matching sliding block was machined with a close sliding fit. This block carries the portable drill and the multi-spindle head so that both move up and down together.

Feed mechanism

A rack was fixed vertically along the column and a pinion gear was mounted on a shaft in the slide region. An operating lever was attached to the pinion shaft so that when the lever is pulled, the slide moves downward to perform drilling, and returns upward by hand or spring.

Multi-spindle head assembly

A solid MS block was drilled and bored to house bearings and spindles. Individual spindles were turned from steel rods, supported in bushes/bearings, and fitted with drill chucks or directly with drill bits. Inside the block, spur/bevel gears were mounted so that rotation from the hand drill chuck is transferred to all spindles.

Mounting of hand drill

The electric hand drill body was clamped to the slide using a bracket, keeping its chuck aligned with the driving gear or input shaft of the multi-spindle head. Alignment was checked so that all spindles are perpendicular to the work support plate.

Assembly and adjustment

The frame, guide, rack-and-pinion, head, and drill were assembled using bolts for easy disassembly. Spindle Centre distances were checked using a measuring scale and trial plates.

Testing and operation

The drill was run without load to verify smooth rotation, absence of excessive vibration, and correct up-down motion of the slide. Test pieces (mild steel or aluminium flat) were clamped on the work plate and multiple holes were drilled in one feed. Time taken and quality of holes (spacing, diameter, finish) were observed and compared with single-spindle drilling.

CHAPTER 6

SELECTION OF MATERIALS

The selection of materials for the multi spindle drilling machine shown in the figure is done to balance strength, rigidity, cost, machinability and ease of fabrication for a student-level project. The main load-carrying structure consists of the base frame and vertical column; these parts are made from welded mild steel (MS) square or rectangular hollow sections and MS flats because mild steel is inexpensive, readily available in Indian workshops, easy to cut and weld, and provides adequate strength and toughness for light and medium duty machine structures. Mild steel sections also allow quick modification of the layout, such as changing the height or adding stiffeners, which is useful during prototype development and academic projects. For industrial machine tools, cast iron is commonly used for beds and columns due to its superior vibration damping and dimensional stability, but it is heavier, more expensive, and requires foundry facilities, so a welded steel frame is more practical for this compact drilling stand where the cutting forces and vibrations are relatively small .

The vertical guide column and any rotating shafts experience bending, torsion and sliding contact, so these parts are selected from medium carbon steels such as EN8 or similar grades. Medium carbon steels have higher tensile strength and hardness than plain mild steel, and can be further improved by heat treatment, which makes them suitable for shafts, guide pillars and spindles that must resist deflection and wear under repeated drilling loads . Using EN8 for the main guide column ensures that the sliding carriage holding the drill and multi-spindle head moves smoothly without permanent bending of the column, even when the operator applies a high manual feed force through the lever. The individual drilling spindles are also made from EN8 or EN19, because these steels possess good fatigue strength and can withstand the torsional stresses produced when

cutting multiple holes simultaneously . The ends of the spindles may be ground for better concentricity so that the drills run true, minimizing run-out and improving hole quality.

The material chosen for the multi-spindle head body is typically a solid block or thick plate of mild steel. Mild steel is preferred here because the head must be machined with several intersecting bores for spindles and gears; MS machines easily, gives a good surface finish, and is strong enough when given sufficient thickness. The head body acts as a rigid housing to maintain accurate centre distance between spindles and to support the bearings or bushes that carry them. For the bearings, two approaches are common: either using sintered bronze or phosphor-bronze bushes, or standard deep-groove ball bearings. Bronze alloys provide good sliding properties and embed small particles, making them suitable where rotational speeds are moderate and lubrication is simple. Ball bearings, on the other hand, offer lower friction and better speed capability, but increase cost; in many student projects, a combination is used, with ball bearings on the more heavily loaded shafts and bushes on lightly loaded ones. In all cases, the bearing material is selected to be harder and more wear-resistant than the shaft surface, but still compatible in terms of friction and lubrication.

Transmission components inside the head, such as spur or bevel gears, must be made from steels that can be hardened, because gear teeth are subjected to repeated contact and bending stresses. Common choices include medium carbon steels that are surface-hardened by carburizing or induction hardening, or alloy steels with good core toughness . Hardened steel gears resist pitting and wear, which is important because any backlash or tooth damage directly affects the synchronization of the spindles and the accuracy of drilled hole spacing. The gear module, face width and material grade are selected according to the transmitted power and speed; for a small hand-drill-driven system, light-duty spur gears with moderate face widths are sufficient, keeping the cost low while

ensuring safe stress levels. Keys and keyways used to lock gears and pulleys to shafts are again cut in EN8 shafts, because this material has adequate shear strength to transmit torque without failure.

The feed mechanism in the photograph is a rack-and-pinion system with a hand lever. The rack is attached to the vertical column or slide, while the pinion is mounted on a shaft supported by the frame; both rack and pinion are manufactured from medium carbon steel for strength and wear resistance. Because these parts operate at low speed but under significant manual load, toughness is more important than high hardness; therefore, they are usually supplied in normalized condition with only moderate surface hardening if required. The lever itself can be made from mild steel flat bar or tube; its length is chosen to provide sufficient mechanical advantage so that the operator can apply the required feed force comfortably. Pivot pins for the lever linkage use medium carbon steel to avoid bending or wear at the joints. Where possible, standard bushes or washers are inserted at pivot points to reduce friction and eliminate play.

Fasteners such as nuts, bolts and washers are typically commercial steel fasteners of property class 4.6 or 8.8, which provide adequate strength for clamping the frame, bearings and attachments. Using standard metric fasteners simplifies assembly and future maintenance, since replacements are easy to obtain. For the work support plate and any fixtures used to hold the job, mild steel plate is chosen because it is strong, easily drilled and tapped, and can be machined to accommodate slots or holes for clamps and bolts. If better wear resistance is needed at clamping surfaces, small hardened steel pads can be inserted in high-pressure regions. The workpiece material itself (mild steel, aluminium or plastic) is selected according to the cutting capability of the hand drill and the required demonstration; softer materials reduce tool wear and are safer for student use .

Finally, the power unit—an off-the-shelf single-phase portable electric drill—is selected primarily for convenience and economy. These drills are designed with robust housings, internal bearings and a standard keyed or keyless chuck, making them ideal as compact spindle drives for small projects. The body is usually made from high-impact plastic with internal aluminium or steel components, combining electrical insulation with structural strength. From a material-selection viewpoint, using a commercial drill avoids the need to design and fabricate a separate motor-spindle unit, reducing complexity and ensuring reliable performance. Electrical cables, switchgear and insulation materials are chosen to meet basic safety standards for the available mains voltage. Overall, the combination of a welded mild steel frame, EN-series steel shafts and spindles, hardened steel gears, suitable bearing materials, standard fasteners and a commercial hand drill results in a cost-effective multi spindle drilling machine that is strong, reasonably accurate, easy to manufacture in a college workshop and adequate for demonstrating multi-hole drilling principles.

CHAPTER 7

WORKING PRINCIPLE

A multi spindle drilling machine operates on the basic drilling principle of converting electrical energy into rotary motion and then into cutting action at the drill tips. The portable electric drill mounted on the carriage supplies rotation to a single driving spindle; this spindle is coupled to a gear train housed in the multi-spindle head, where spur or bevel gears divide the input motion and transmit it to several parallel spindles. As a result, all the drills mounted in these spindles rotate simultaneously at nearly the same speed and are ready to cut multiple holes at once.

Feeding of the drills into the workpiece is achieved by a mechanical rack-and-pinion system controlled by a hand lever. The head and drill are fixed to a sliding carriage that moves along a vertical guide column; a straight rack is attached to this column and meshes with a pinion mounted on a shaft operated by the lever. When the operator pulls the lever downward, the pinion turns and walks along the rack, converting the lever's rotary motion into linear downward movement of the carriage, so that all rotating drills advance together into the clamped workpiece.

Because all the spindles are rigidly fixed in the head with predetermined centre distances, the holes produced in a single stroke have accurate and constant spacing. Once the required depth is reached, the operator releases or lifts the lever, causing the carriage to move upward and retracting all drills simultaneously. This synchronized rotation and feed allow several holes to be drilled in one operation, reducing cycle time and improving positional accuracy compared with drilling one hole at a time on a conventional single-spindle machine.

CHAPTER 8

PROCESS INVOLVED IN FABRICATION

The fabrication of this multi spindle drilling machine involves a sequence of basic workshop processes starting from raw material preparation and ending in assembly and testing. First, design dimensions are finalized and mild-steel square tubes, flats and plates for the base and vertical frame are marked and cut to size using power saw or hacksaw; edges are ground and cleaned, then the parts are tack-welded on a flat surface to maintain squareness and finally fully welded to form the base, uprights, and diagonal stiffeners. Next, the guide arrangement and support brackets are made: the vertical guide column and support members are cut and welded, and brackets for mounting the rack, pinion shaft, gear housing and drill holder are fitted; holes are drilled in the frame for bolts, and surfaces in sliding contact are ground or filed to obtain smooth motion.

In the next stage, machining operations are performed on rotating and sliding parts. Medium-carbon steel rods for spindles, shafts and guide pillars are turned on the lathe to required diameters, shoulders and threads; keyways are milled or slotted where gears or pulleys are to be mounted. The multi-spindle head body or drill holder block is cut from thick plate or solid bar and machined on the drilling machine or milling machine to create accurately spaced holes for spindles, bearings and fasteners. Ready-made spur gears or flame-cut gear blanks are finished and bored on the lathe, then keyed to their shafts to form the gear train visible in your photo. Bushes or bearings are pressed into the head and into frame brackets, and the rack is either purchased as a standard item or cut and then welded/bolted to the column.

Finally, all parts are assembled and the machine is finished. The frame is cleaned of welding slag, checked for alignment and repainted to prevent corrosion. Shafts, gears, and spindles are assembled with proper lubrication; the rack-and-pinion and lever mechanism are fitted and adjusted so that the sliding carriage moves freely along the guide without play. The portable electric drill or drilling unit is clamped to the carriage, aligned with the input shaft of the multi-spindle head, and the head itself is bolted at the correct height above the work area. After tightening all fasteners, the machine is test-run without load to check for vibration or interference, then trial holes are drilled in scrap workpieces to verify that all spindles rotate correctly, hole spacing is accurate, and the feed operates smoothly; any misalignment or excessive noise detected during these trials is corrected by re-shimming, re-welding or re-machining the affected parts.

CHAPTER 9

MODELING (2D & 3D)

9.1 2D Model:

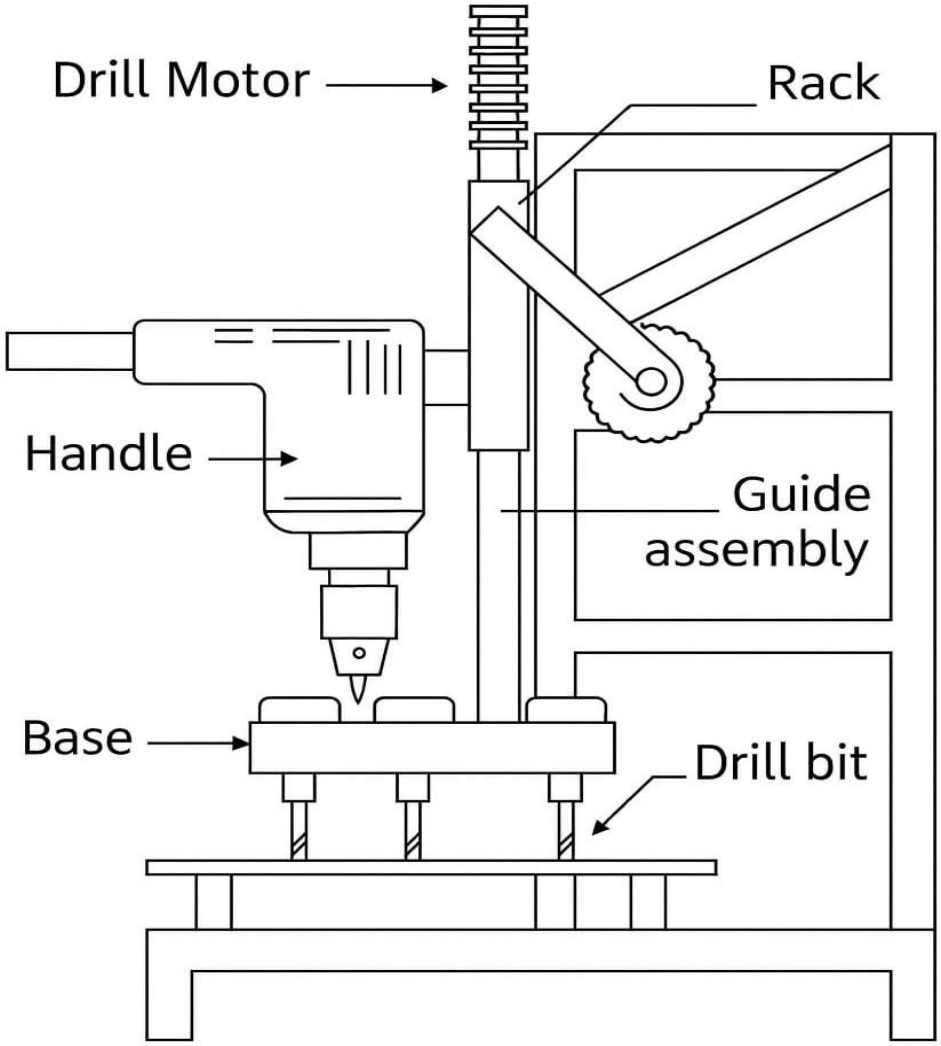
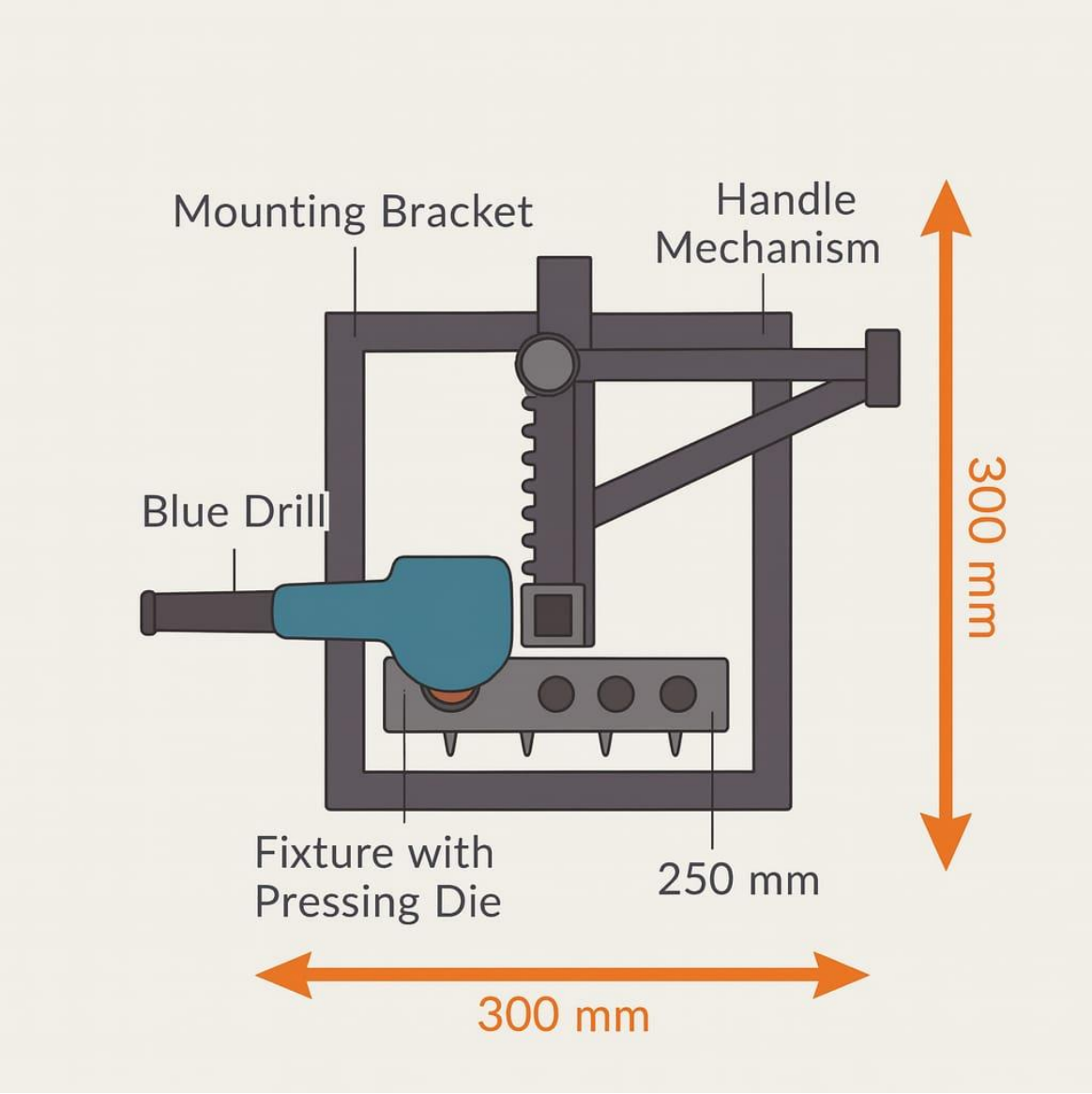


Fig: Front view of Multi spindle drilling machine



Top View

Fig: Top view of Multi spindle drilling machine

9.2 3D Model:

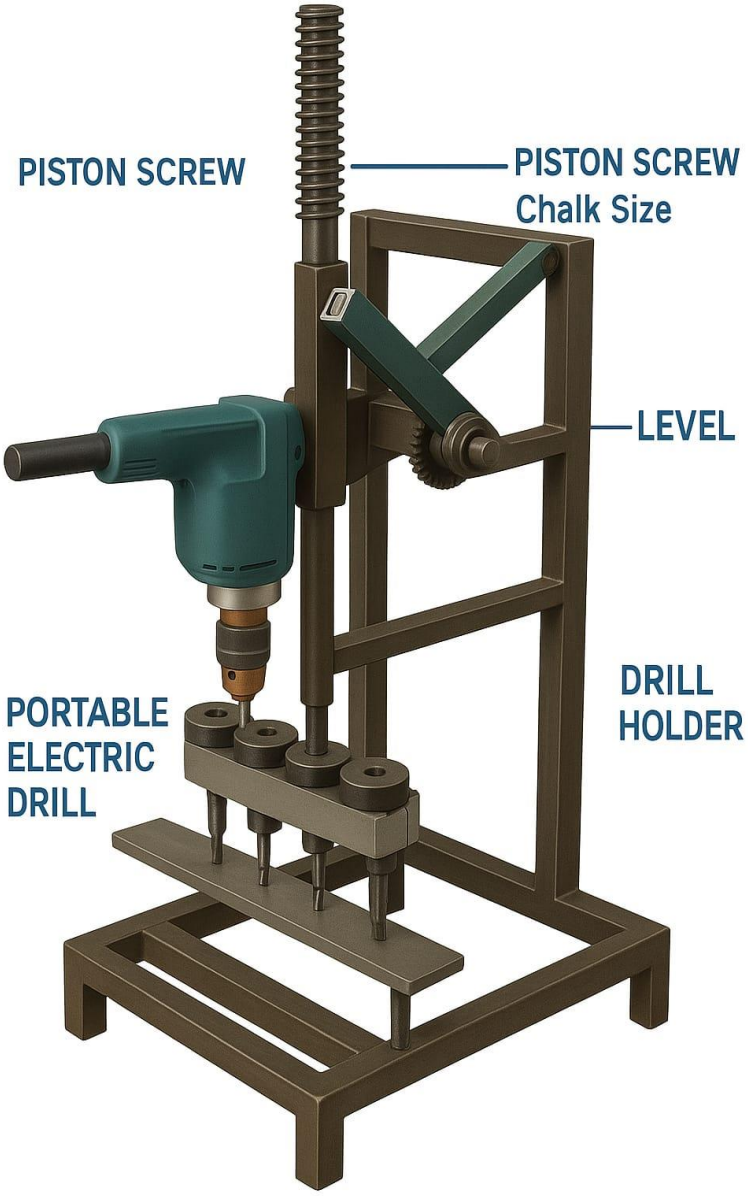


Fig 9.3

3D model of Multi spindle drilling machine

CHAPTER 10

BUDGET

10.1 Budget:

Table 10.1

S.NO	TESTING	ACTUAL COST
1	STEEL FRAME & BASE	2000
2	PLATES AND GUIDE COLUMN	1200
3	RACK, PINION AND GEARS	1500
4	BEARINGS	1000
5	ELECTRIC DRILL	2800
6	MULTI SPINDLE HEAD BLOCK MACHININIG	1500
7	SMALL HARDWARES	700
8	MACHINING CHARGES	800

9	PAINT & SWITCH	500
	TOTAL	**12000**

CHAPTER 11

FINAL VIEWS OF PROJECT



FIG 11.1

Front view of Multi spindle drilling machine

CHAPTER 12

CONCLUSION

A multi spindle drilling machine project demonstrates that significant gains in productivity and accuracy can be achieved with a relatively simple and low-cost mechanism. By transmitting power from a single drive to several spindles, the machine drills multiple holes in one feed stroke, which reduces cycle time, labour effort and cost per component compared with a conventional single-spindle drill. Because the spindles are rigidly located in a common head, the spacing between holes remains constant, improving positional accuracy and repeatability, which is essential for components such as brackets, flanges and small fabrication parts.

For an academic fabrication like yours, using a welded mild-steel frame, rack-and-pinion feed and a portable electric drill as the power source makes the setup economical while still providing a clear practical understanding of gear trains, power transmission and multi-tool operation. The project also shows that a simple attachment or stand can convert an ordinary hand drill into a special-purpose machine suitable for small-scale batch production, giving students experience in designing for manufacturability, safety and maintainability. Overall, the multi spindle drilling machine is a useful teaching model as well as a functional workshop tool that illustrates how thoughtful mechanical design can simultaneously improve speed, quality and cost effectiveness in manufacturing.

CHAPTER 13

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